

PSUEDOCODES

Below is the pseudocode for the surface blasting design calculator.

Start

Declare variables:BD, M, d, , BH, PFtech, SL, SD, B, S, PFact

INPUT BD

INPUT D

INPUT S/B

INPUT BH

Calculate

$$M = (d \times BD^2) / 1273$$

IF the blast has environmental issues THEN

$$SL = 30 \times BD$$

ELSE

$$SL = 20 \times BD$$

END IF

OUTPUT SL

Calculate the Technical Burden

$$B = \sqrt{(M \times (BH - SL)) / (S/B \times PF_{tech} \times BH)}$$

OUTPUT B

Calculate

$$SD = (1/4) \times B$$

OUTPUT SD

IF the rock type is hard rock THEN

$$PFact = M \times (BH - SL + SD) / (B \times S \times BH)$$

ELSE

$$PFact = M \times (BH - SL - SD) / (B \times S \times BH)$$

END IF

OUTPUT PFact

End

Pseudocode for mean rock fragmentation

START

INPUT A, REE, PF, p, K, D, BH, SL

Calculate $M = (p \times D^2) / 1273$

Calculate $Q = M \times (BH - SL)$

FUNCTION MeanFragmentation

$$\text{MeanFragmentationSize} = A \times (K^{-0.8}) \times (Q^{0.167}) \times ((115 / \text{REE})^{0.633})$$

RETURN MeanFragmentationSize

END FUNCTION

OUTPUT MeanFragmentationsize

END